Mathematical and Computational Methods

Exercise Sheet 7: Differentiation I: limits and the rules

Tutorial route (▶, in this order, about 80 minutes): 1.2(b,c), 2.6, 2.7, 2.8, 2.9, 2.4(b), 2.1(a). Everything else is for home study.

Levels: ⋆ drill, ⋆⋆ core, ⋆⋆⋆ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Limits, continuity

Exercise 1.1 *(⋆ drill limits at a point).* Evaluate, cancelling a common factor or rationalising first where needed:

(a) $\lim_{x\to 2} \dfrac{x^2-4}{x-2}$;

(b) $\lim_{x\to 3} \dfrac{x^2-9}{x^2-x-6}$;

(c) $\lim_{x\to 0} \dfrac{\sqrt{x+1}-1}{x}$;

(d) $\lim_{x\to 0} \dfrac{\sin 5x}{x}$.

Exercise 1.2 *(▶ ⋆ drill one-sided limits and asymptotes).* (a) Find $\lim_{x\to 0^+} \dfrac{1}{x}$ and $\lim_{x\to 0^-} \dfrac{1}{x}$, and state the vertical asymptote.

(b) For $f(x) = \begin{cases} x^2, & x < 1 \\ 2x, & x \geq 1 \end{cases}$ find the one-sided limits at $x = 1$ and decide whether $\lim_{x\to 1} f(x)$ exists.

(c) Divide by the highest power of x in the denominator to find $\lim_{x\to\infty} \dfrac{2x+1}{x^2+3}$ and $\lim_{x\to\infty} \dfrac{4x^3-x}{2x^3+7}$, and state each horizontal asymptote $y = L$.

Exercise 1.3 *(⋆ drill continuity).* Find the constant that makes each function continuous everywhere.

(a) $f(x) = \dfrac{x^2-1}{x-1}$ for $x \neq 1$, and $f(1) = k$.

(b) $g(x) = \begin{cases} cx+1, & x \leq 2 \\ x^2, & x > 2. \end{cases}$

The derivative; standard derivatives and rules

Exercise 1.4 *(⋆ drill first principles)*. Using $f'(x) = \lim_{\delta x \to 0} \dfrac{f(x+\delta x) - f(x)}{\delta x}$, differentiate from first principles: (a) $f(x) = x^2 - 3x$; (b) $f(x) = \dfrac{1}{x}$.

Exercise 1.5 *(⋆ drill standard derivatives, successive derivatives)*. Write down the derivative of each (use a power where helpful):

$$\text{(a) } x^5, \quad \text{(b) } \sqrt{x}, \quad \text{(c) } \frac{1}{x^3}, \quad \text{(d) } \mathrm{e}^{2x}, \quad \text{(e) } \ln x,$$
$$\text{(f) } \sin 3x, \quad \text{(g) } \cos\tfrac{x}{2}, \quad \text{(h) } \tan x, \quad \text{(i) } \cosh x.$$

(j) For $y = x^4$, list y', y'', y''', $y^{(4)}$, $y^{(5)}$.

Exercise 1.6 *(⋆ drill product and quotient rules)*. Name u and v each time. Product rule $(uv)' = u'v + uv'$: (a) $x^2\mathrm{e}^x$; (b) $x \sin x$. Quotient rule $\left(\frac{u}{v}\right)' = \dfrac{u'v - uv'}{v^2}$: (c) $\dfrac{x}{x+1}$; (d) $\dfrac{\sin x}{x}$.

Exercise 1.7 *(⋆ drill chain rule)*. Differentiate: (a) $(3x+1)^5$; (b) e^{x^2}; (c) $\sin(x^2)$; (d) $\ln(1+x^2)$.

Further techniques

Exercise 1.8 *(⋆ drill parametric, implicit, logarithmic drill)*. Find $\frac{\mathrm{d}y}{\mathrm{d}x}$: (a) parametrically, for $x = \cos t$, $y = \sin t$; (b) implicitly, for $xy = 1$; (c) logarithmically, for $y = x^x$, $x > 0$.

2 Tutorial problems

Exercise 2.1 *(► ⋆⋆ core continuity, differentiability and a corner)*. Both parts turn on the one-sided limits of the difference quotient at the join.

(a) Let $f(x) = \begin{cases} x^2, & x \le 1 \\ ax + b, & x > 1. \end{cases}$ Find a, b so that f is *differentiable* at $x = 1$. Why is your f automatically continuous there?

(b) Using the one-sided limits of the difference quotient at 0, show that $\mathrm{ReLU}(x) = \max(0, x)$ is continuous at 0 but not differentiable there.

Exercise 2.2 *(⋆⋆ core first principles and the tangent (geometric reading))*. Let $f(x) = \sqrt{x}$.

(a) Show from first principles that $f'(x) = \dfrac{1}{2\sqrt{x}}$.

(b) Hence find the equation of the tangent line to $y = \sqrt{x}$ at $x = 4$.

Exercise 2.3 *(⋆⋆ core velocity and acceleration — physics)*. A particle moves along a line with displacement $s(t) = t^3 - 6t^2 + 9t$ metres, $t \ge 0$ seconds.

(a) Find the velocity $v(t) = \dot{s}$ and acceleration $a(t) = \ddot{s}$.

(b) When is the particle momentarily at rest?

(c) In which direction is it moving at $t = 2$?

(d) Give the acceleration at each rest instant and interpret its sign.

Exercise 2.4 *(▶ ⋆⋆ core sensitivity, the sigmoid and a learning curve — data science).* (a) The logistic function is $\sigma(x) = \dfrac{1}{1+\mathrm{e}^{-x}}$. Show, using the chain rule, that $\sigma'(x) = \sigma(x)\big(1-\sigma(x)\big)$.

(b) A model's error is $L(w) = (w-3)^2 + 1$. Compute $\frac{\mathrm{d}L}{\mathrm{d}w}$ and evaluate it at $w = 5$. Reading the derivative as a *sensitivity*, should a *small* gradient-descent step increase or decrease w? Write out the gradient-descent update $w \leftarrow w - \eta\,\frac{\mathrm{d}L}{\mathrm{d}w}$ (learning rate $\eta > 0$) for this L. (Unit 8 develops the method, including how large η may be.)

(c) During training a loss decays as $\ell(t) = \mathrm{e}^{-t/5}$, where t is the epoch number. Find the rate of change $\ell'(t)$, evaluate it at $t = 0$ and $t = 5$, and say what the two values tell you about training.

Exercise 2.5 *(⋆⋆ core spot the error — conceptual).* Each line contains one mistake. Identify it and give the correct derivative.

(a) "$\frac{\mathrm{d}}{\mathrm{d}x}(x^2+1)^3 = 3(x^2+1)^2$."

(b) "$\frac{\mathrm{d}}{\mathrm{d}x}\big(x^2\mathrm{e}^x\big) = 2x\,\mathrm{e}^x$."

(c) "$\frac{\mathrm{d}}{\mathrm{d}x}\big(\sin x\cos x\big) = \cos x\,(-\sin x)$."

Exercise 2.6 *(▶ ⋆⋆ core quotient-then-chain (worked-example anchor)).* Differentiate $y = \dfrac{(2x-1)^4}{x^2+1}$, simplifying as far as possible.

Exercise 2.7 *(▶ ⋆⋆ core implicit differentiation (worked-example anchor)).* The folium of Descartes is $x^3 + y^3 = 6xy$.

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ by implicit differentiation.

(b) Hence find the slope of the curve at the point $(3, 3)$.

Exercise 2.8 *(▶ ⋆⋆ core parametric tangents).* A curve is given by $x = t^2,\ y = t^3 - 3t$.

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of t.

(b) Find the point(s) with a horizontal tangent and the point(s) with a vertical tangent.

Exercise 2.9 *(▶ ⋆⋆ core logarithmic differentiation — a multi-factor expression).* Use logarithmic differentiation to differentiate $y = (x^2+1)^3(2x-1)^4\,\mathrm{e}^{-5x}$.

3 Further practice (home)

Exercise 3.1 *(⋆⋆ core classify the breaks).* For each function, find every point where the graph breaks (the function is undefined there, or is defined but not continuous) and classify the break as *removable* (the two one-sided limits exist and agree, but the function is undefined there or takes a different value), a *jump* (both one-sided limits exist but differ), or *infinite* (a one-sided limit is $\pm\infty$, so the graph has a vertical asymptote).

$$\text{(a) } f(x) = \frac{x-2}{x^2-4}, \qquad \text{(b) } g(x) = \frac{1}{x-1}, \qquad \text{(c) } h(x) = \begin{cases} x, & x < 0 \\ x+2, & x \geq 0. \end{cases}$$

Exercise 3.2 *(⋆⋆ core true or false — conceptual).* State whether each is true or false, with a one-line reason.

(i) If f is continuous at a then f is differentiable at a.

(ii) $(uv)' = u'v'$.

(iii) For $\lim_{x\to a} f(x)$ to exist, $f(a)$ must be defined.

(iv) If $\lim_{x\to\infty} f(x) = L$ then $y = L$ is a horizontal asymptote.

(v) $\frac{\mathrm{d}}{\mathrm{d}x}x^n = nx^{n-1}$ holds only for integer n.

(vi) $f(x) = x^{1/3}$ is differentiable at $x = 0$.

Exercise 3.3 *(⋆⋆ core the differentiation bank)*. Differentiate (state the rule used):

(a) $\mathrm{e}^{x^2}\sin x$, (b) $x^2\ln x$, (c) $\dfrac{x}{\sqrt{x^2+1}}$, (d) $\mathrm{e}^{-x}\cos 2x$, (e) $\dfrac{\ln x}{x}$, (f) $\sqrt{1+\sqrt{x}}$ $(x>0)$.

Exercise 3.4 *(⋆⋆ core parametric bank, including a second derivative)*. (a) For $x = 2t,\ y = t^2 - 1$ find $\frac{\mathrm{d}y}{\mathrm{d}x}$ and $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$.

(b) For the circle $x = r\cos t,\ y = r\sin t$ find $\frac{\mathrm{d}y}{\mathrm{d}x}$ and $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$.

Exercise 3.5 *(⋆⋆ core implicit bank, including a second derivative)*. (a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for $x^2 - xy + y^2 = 7$.

(b) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for $\sin(xy) = x$.

(c) For the hyperbola $x^2 - y^2 = 16$, find $\frac{\mathrm{d}y}{\mathrm{d}x}$ and then $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$, expressing the second derivative purely in terms of y.

Exercise 3.6 *(⋆⋆ core logarithmic differentiation of a quotient)*. Use logarithmic differentiation to differentiate $y = \dfrac{\sqrt{x^2+1}\,\mathrm{e}^x}{(x+3)^4}$.

Exercise 3.7 *(⋆ ⋆ ⋆ challenge harder limits)*. Evaluate, quoting $\lim_{u\to 0}\frac{\sin u}{u} = 1$ where needed:

$$\text{(a)}\ \lim_{x\to 0}\frac{1-\cos x}{x^2},\qquad \text{(b)}\ \lim_{x\to\infty}\left(\sqrt{x^2+x}-x\right),\qquad \text{(c)}\ \lim_{x\to 0}\frac{\sin 3x}{\sin 5x}.$$

Exercise 3.8 *(⋆ ⋆ ⋆ challenge the cycloid)*. A cycloid is traced by $x = a(t-\sin t),\ y = a(1-\cos t)$ $(a>0,\ t\geq 0)$. Show that $\frac{\mathrm{d}y}{\mathrm{d}x} = \cot\frac{t}{2}$, and state the values of t at which the tangent is horizontal.

Exercise 3.9 *(⋆ ⋆ ⋆ challenge a tower of powers)*. Differentiate $y = x^{x^x}$ for $x > 0$. *Hint: take logs twice, or differentiate x^x first.*